



AI Assignment-3

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1. Introduction:

This report outlines the implementation of a simplified barcode reader system, which consists of two main components: an encoder and a decoder. The encoder generates a barcode image from a given string, while the decoder reads and reconstructs the string from the generated barcode image. This task showcases the application of computer vision techniques in processing and interpreting barcodes, which are widely used in product packaging for efficient data retrieval.

2. Simplified System Simulation using Computer Vision

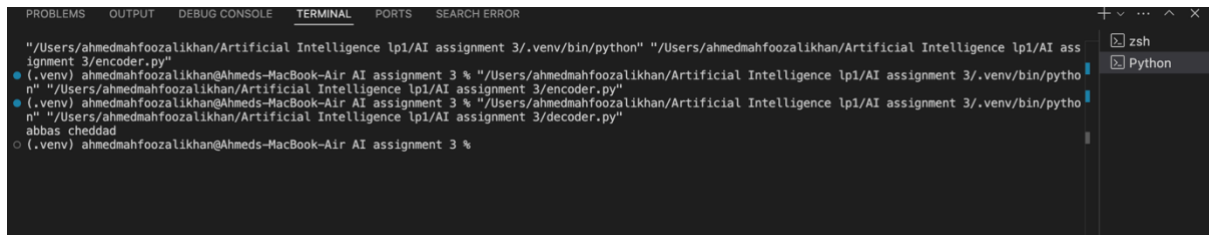
The task involves designing and implementing a system that encodes a string into a barcode-like image and then decodes that image back into the original string. The system is simplified in terms of how barcodes represent characters, focusing on creating and reading custom bar widths based on the position of letters in the alphabet

2.1 Encoder

- Step between bars: 9 pixels
- Bar height: Fixed height from row 10 to row 350
- Shorter bars to represent spaces: From row 150 to row 250
- Each character's width corresponds to its position in the alphabet plus one pixel.
- The character 'A' or 'a' has a width of 2 pixels, while 'S' or 's' has a width of 20 pixels.
- **Image Saving:** The generated barcode image is saved as output.png

2.2 Decoder

- **Image Reading:** The output.png image is loaded for processing.
- **Middle Row Scanning:** The algorithm scans the middle row (row 200) of the image to identify bar widths.
- **String Reconstruction:** The widths of the bars are calculated, and corresponding characters are appended to the string.
- **Output:** The string decoded has been shown below

A screenshot of a terminal window with a dark background. The terminal shows a series of commands and their outputs. The first command is a full path to a Python script. The second command is a relative path to the same script. The third command is a relative path to the script. The fourth command is a relative path to the script. The output of the fourth command is a string. The terminal window has tabs for 'zsh' and 'Python'. The top of the window shows tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL', 'PORTS', and 'SEARCH ERROR'.

```
"/Users/ahmedmahfoozalikhah/Artificial Intelligence lp1/AI assignment 3/.venv/bin/python" "/Users/ahmedmahfoozalikhah/Artificial Intelligence lp1/AI assignment 3/encoder.py"
(.venv) ahmedmahfoozalikhah@Ahmeds-MacBook-Air AI assignment 3 % "/Users/ahmedmahfoozalikhah/Artificial Intelligence lp1/AI assignment 3/.venv/bin/python"
n" "/Users/ahmedmahfoozalikhah/Artificial Intelligence lp1/AI assignment 3/encoder.py"
(.venv) ahmedmahfoozalikhah@Ahmeds-MacBook-Air AI assignment 3 % "/Users/ahmedmahfoozalikhah/Artificial Intelligence lp1/AI assignment 3/.venv/bin/python"
n" "/Users/ahmedmahfoozalikhah/Artificial Intelligence lp1/AI assignment 3/decoder.py"
abbas cheddad
(.venv) ahmedmahfoozalikhah@Ahmeds-MacBook-Air AI assignment 3 %
```

3. Conclusion

The simplified barcode reader system successfully encodes a string into a barcode and decodes it back into the original string. The implementation utilized Python's PIL library for image creation and manipulation, demonstrating the practical application of computer vision techniques in handling barcodes.

4. Submission

The following files are submitted:

- encoder.py: Contains the code for encoding the string into a barcode.
- decoder.py: Contains the code for decoding the barcode back into a string.
- output.png: The generated barcode image representing the input string.

5. References

1. <https://simple.wikipedia.org/wiki/Barcode>
2. <https://en.wikipedia.org/wiki/Barcode>
3. Refer class notes
4. Use ChatGPT for gathering information: <https://chatgpt.com/share/671399ce-abb4-8008-ab90-ee41fb872c77>